Generation as Computation

Generalised Computation in the Set-theoretic Universe and Beyond
(Thesis Proposal)

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Outline

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Gödel's Informal Proposition

- In an essay Gödel wrote in 1958 (but only published posthumously) [4], he criticised Hilbert's choice of "intuitively clear" notions meant to make up the bedrock of classical mathematics.
- He lamented that "the domain of 'finitary mathematics"' is unnaturally restrictive.
- In the same work, Gödel informally described a notion of infinitary computation built inductively from natural numbers.
- He called this notion *computable functions of finite type*.

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In Classical Computability Theory

- "A program is a formula (of a particular form)" — an oft-repeated slogan.
- An oracle machine can functionally be represented as a sufficiently simple parametrised formula in the language of arithmetic.
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- α-recursion theory, born from ideas by Takeuti [1], Kreisel and Sacks [2] and some others, in the 1960s.
- E-recursion theory, formulated by Normann [3] in the 1970s.
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1. Introduction
   - Background: Infinitary Computation
   - Background: Formulas as Programs
   - Overview
2. Generalised Computation Within V
   - Sequential Algorithms
   - Generalised Sequential Algorithms
   - Relative Computability
3. Generalised Computation Beyond V
   - Degrees of Small Extensions
   - Local Method Definitions

In Greater Generality...

- Relative computability is but a form of parametrised definability over some structure.
- Vanilla computability is a special case whereby the parameter in question contains the least amount of information (e.g. using the empty set as parameter).
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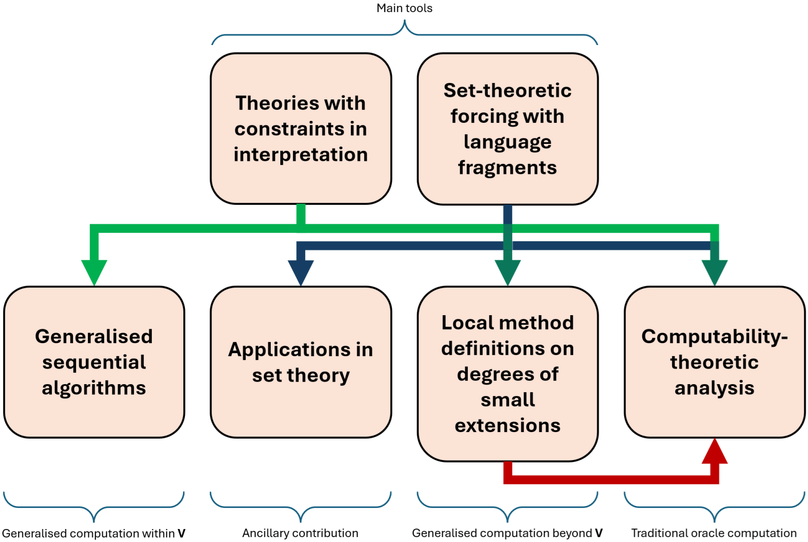
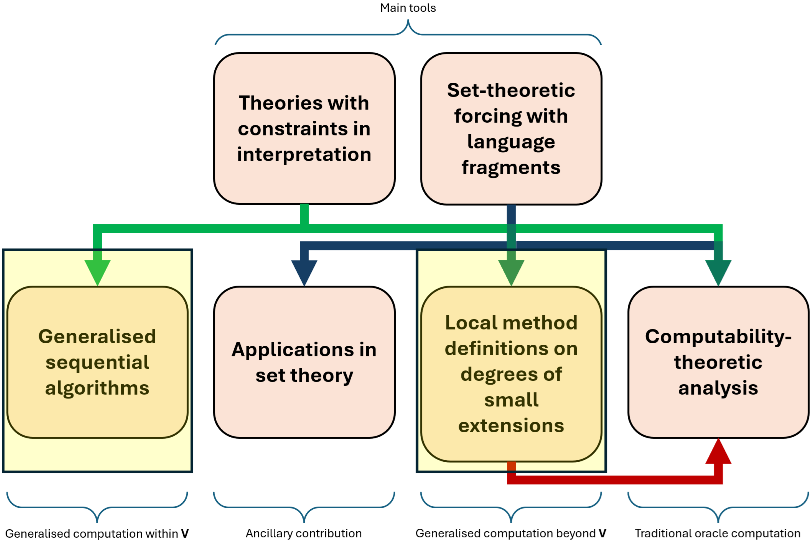
Overview Chart



Today's Focus



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Higher-Level Descriptions

- The explicit models of computation mentioned so far (BSSMs, ITTMs, etc.) are rather low-level and relatively concrete.
- Computer scientists were interested in more inclusive descriptions of algorithms.
- We want to be able to describe programs in pseudocode, instead of being bound to a fixed programming language.
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Gurevich's formulation involves proper classes, but it can be adapted to avoid size issues.

Definition

- A *sequential algorithm* comprises a set of *states*, including *initial states*, and a *transition function* mapping one state to another.
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Generality and Intuitiveness

- States are analogous to Turing machine configurations.
- An initial state corresponds to a starting configuration upon receiving an input.
- A final state corresponds to an end configuration from which the output can be read.
- A transition function parallels that of a Turing machine: it codes how the machine behaves.
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- It seems like the go-to candidate for generalisation, if we want an flexible and high-level model of infinitary computation.
- Two principles, isolated by Sieg in [8] as integral to classical models of computation, guide our search for the "correct" infinitary version of sequential algorithms.
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Redefining a Transition Function

- We want to talk about local features between two states.
- Since they are first-order structures with the same signature and base set, local features are synonymous with properties given through the truth predicate $\models$.
- We modify the predicate into $\models_2$ so that we can talk about any two such structures simultaneously.
- Every *generalised sequential algorithm* (*GSeqA*) has its transitions governed by a transition formula: transition from s_1 to s_2 iff $(s_1, s_2) \models_2 \phi$ for transition formula ϕ.

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Ordinal Base Sets

- To shift the burden of programming to the definition of the transition formula, we streamline the definitions of other components of a GSeqA.
- We specify the base set of any state to be a limit ordinal.
- More precisely, every GSeqA $\mathfrak{M}$ has an associated limit ordinal $\kappa(\mathfrak{M})$.
- Every state of a GSeqA $\mathfrak{M}$ is an expansion of the structure $(\kappa(\mathfrak{M}); \in)$ over a fixed signature $\sigma(\mathfrak{M})$ associated with $\mathfrak{M}$.
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- The base set being an ordinal also fits the basic intuition of sequential memory when one thinks about computation.

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Ordinal Base Sets

- To shift the burden of programming to the definition of the transition formula, we streamline the definitions of other components of a GSeqA.
- We specify the base set of any state to be a limit ordinal.
- More precisely, every GSeqA $\mathfrak{M}$ has an associated limit ordinal $\kappa(\mathfrak{M})$.
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- For technical purposes, we focus on the subclass of GSeqAs with ordinal parameters, called *generalised sequential algorithm with parameters* (*GSeqAP*s).
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Input/Output Paradigm

- The interpretation of "computing B from A" is implicit in a sequential algorithm.
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Transition Formulas

- Let $\mathfrak{M}$ be a GSeqAP.
- For each declared variable $\ulcorner X \urcorner \in \sigma(\mathfrak{M})$, we associate it with a Δ_0 formula $\phi^{\ulcorner X \urcorner}$ of some specific form over $\sigma(\mathfrak{M})$.
- $\phi^{\ulcorner X \urcorner}$ specifies "pointwise" the next state's interpretation of $\ulcorner X \urcorner$ based on the current state's interpretation.
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Initial States

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- For each declared variable $\ulcorner X \urcorner \in \sigma(\mathfrak{M})$, we associate it with a Δ_0 formula $\psi^{\ulcorner X \urcorner}$ of some specific form over $\in$.
- $\psi^{\ulcorner X \urcorner}$ ought to specify "pointwise" the unique starting behaviour of $\ulcorner X \urcorner$.
- Then $\phi^d_{\mathfrak{M}}$ of $\mathfrak{M}$, defined to be the conjunction of the $\psi^{\ulcorner X \urcorner}$s, determines the starting behaviour of all declared variables.
- If $A \subset \kappa(\mathfrak{M})$, then there is a unique initial state s of $\mathfrak{M}$ such that s interprets In as A, Out as $\emptyset$ and $s \models \phi^d_{\mathfrak{M}}$ of $\mathfrak{M}$.
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GSeqAP: Definition

- A GSeqAP $\mathfrak{M}$ is a tuple

$$(\kappa(\mathfrak{M}), \sigma(\mathfrak{M}), \mathrm{p}(\mathfrak{M}), \phi^{\tau}_{\mathfrak{M}}, \phi^{d}_{\mathfrak{M}})$$

  satisfying the properties we have mentioned.
- Let $S(\mathfrak{M})$ denote the set of states of $\mathfrak{M}$, derivable from $\kappa(\mathfrak{M})$, $\sigma(\mathfrak{M})$ and $\mathrm{p}(\mathfrak{M})$.
- The transition (partial) function $\tau_{\mathfrak{M}}$ of $\mathfrak{M}$ can then be derived from $S(\mathfrak{M})$ and $\phi^{\tau}_{\mathfrak{M}}$ as follows:

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Outline

A Call for Transfinite Runs

- *Runs* (or *computations*) are presumed to be finite in a sequential algorithm.
- Even with our restrictions, one can easily compress a finite run into a single step.
- If we allow transfinite inputs and outputs, it makes sense to allow transfinite runs.
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Taking Limits

- We want the definition of a limit state to depend locally on the states that come before it in a run.
- This can again be formalised using the first-order truth predicate in a "pointwise" Δ_0 way.
- Under this constraint, we show that with enough space (increasing $\kappa(\mathfrak{M})$ if necessary), taking limit/limsup/liminf wherever possible "pointwise" can simulate all other definitions of a limit state.

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Terminating Runs

Definition

A *terminating run* of a GSeqAP $\mathfrak{M}$ is a function from some successor ordinal δ into $S(\mathfrak{M})$ such that

- $Sq(0)$ is an initial state,
- $Sq(\alpha+1) = \tau_{\mathfrak{M}}(Sq(\alpha)) \neq Sq(\alpha)$ for all $\alpha < \delta - 1$,
- $Sq(\gamma)$ is obtained from $Sq \restriction \gamma$ "pointwise" by taking liminf if possible and resetting to 0 otherwise, for limit ordinals $\gamma < \delta$, and
- $\tau_{\mathfrak{M}}(Sq(\delta-1)) = Sq(\delta-1)$.

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- $Sq(0)$ is an initial state,
- $Sq(\alpha+1) = \tau_{\mathfrak{M}}(Sq(\alpha)) \neq Sq(\alpha)$ for all $\alpha < \delta - 1$,
- $Sq(\gamma)$ is obtained from $Sq \restriction \gamma$ "pointwise" by taking liminf if possible and resetting to 0 otherwise, for limit ordinals $\gamma < \delta$, and
- $\tau_{\mathfrak{M}}(Sq(\delta-1)) = Sq(\delta-1)$.

We call δ the *length* of Sq.

Relative Computability Relations

- A GSeqAP computes B from A iff it has a run that starts with In interpreted as A and terminates with Out interpreted as B.
- $B \leq^P_\kappa A$ iff some GSeqAP $\mathfrak{M}$ with $\kappa(\mathfrak{M}) = \kappa$ computes B from A.
- $B \leq^{P,s}_\kappa A$ iff some GSeqAP $\mathfrak{M}$ with $\kappa(\mathfrak{M}) = \kappa$ computes B from A in a terminating run of length $< \kappa$.
- $B \leq^P A$ iff $B \leq^P_\kappa A$ for some limit ordinal κ.
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Degrees of Constructibility

- We show that $B \leq^P A$ iff $B \in L[A]$, where $L[A]$ is Gödel's constructible universe relative to A.
- This means the degrees of relative computability derived from $\leq^P$ are exactly the degrees of constructibility.
- The very restricted (at first glance) programming we allow in GSeqAPs actually has the full power of transfinite recursion.
- In fact, the above is already true for the much lower-level α-machines in Koepke's theory of α-computability (see e.g. [7]).

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The Relations $\leq_{\kappa}^{P,s}$ and $\leq_{\kappa}^{P}$

- An ordinal κ is *admissible* iff $(L_{\kappa}; \in)$ is a model of Kripke-Platek set theory.
- When κ is admissible, $\leq_{\kappa}^{P,s}$ and $\leq_{\kappa}^{P}$ are both defined.
- It can be proven that if κ_1 and κ_2 are admissible, $\kappa_1 < \kappa_2$ and $B \leq_{\kappa_1}^{P} A$ (resp. $B \leq_{\kappa_1}^{P,s} A$), then $B \leq_{\kappa_2}^{P} A$ (resp. $B \leq_{\kappa_2}^{P,s} A$).
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Summary Table

In the following table we compare various relative computability relations when they are restricted to subsets of admissible ordinals α.

Property \ Relation	$\leq_\alpha$	$\preceq_\alpha$	$\leq_\alpha^P$	$\leq_\alpha^{P,s}$
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Outline

1. Introduction
   - Background: Infinitary Computation
   - Background: Formulas as Programs
   - Overview
2. Generalised Computation Within V
   - Sequential Algorithms
   - Generalised Sequential Algorithms
   - Relative Computability
3. Generalised Computation Beyond V
   - Degrees of Small Extensions
   - Local Method Definitions

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- What do we mean by "sets outside V"?
- Step out of V and treat V as a countable transitive model of ZFC (henceforth denoted CTM).
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Outer Models

- Let U_1 and U_2 be CTMs. U_2 is an outer model of U_1 iff
  - $U_1 \subset U_2$, and
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Definition

Let V be a CTM. The *outward multiverse centred at V* is the set

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Due to Jensen's result on "coding the universe", we have the following theorem.

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Given a CTM V, $(\mathbf{M}_S(V), \subset)$ is a cofinal subposet of $(\mathbf{M}(V), \subset)$.

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The next proposition is easy to see.

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- By the previous proposition, it suffices to consider equivalence classes arising from the natural reducibility relation on

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Degrees over V

- Specifically, define the binary relation $\leq^S$ on $\mathscr{G}(V)$ by

$$x \leq^S y \iff V[x] \subset V[y].$$

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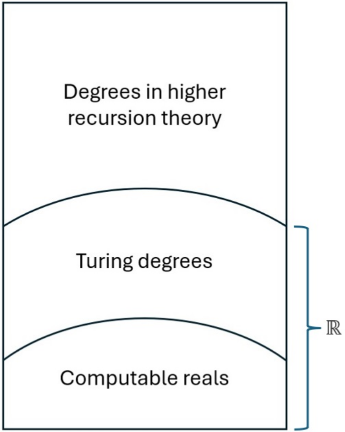

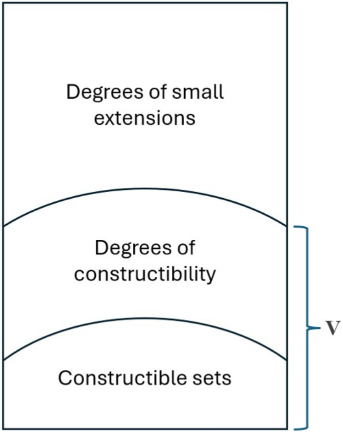

Figure: comparison between conventional notions of relative computability (left) and our generalised notions (right).

Multiverse of Generic Extensions

Definition

Let V be a CTM. The *outward generic multiverse centred at* V is the set

$$\mathbf{M}_F(V) := \{W : W \text{ is a forcing extension of } V\}.$$

The following fact is basic.

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Every generic extension is a small extension. Equivalently,

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$\mathbf{M}_F(V)$ *is downward-closed in* $\mathbf{M}(V)$*, and thus also in* $\mathbf{M}_S(V)$.

On the other hand, the next fact can be derived from arguments using class forcing.

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Given a CTM V, $(\mathbf{M}_S(V) \setminus \mathbf{M}_F(V), \subset)$ *is a cofinal subposet of* $(\mathbf{M}_S(V), \subset)$.

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Outline

Referring to Small Extensions in V

- Often it is useful to refer to small extensions of V within V.
- Such references in general cannot isolate any non-trivial small extension.
- Think of them as descriptions in V picking out sets of small extensions of V, when evaluated outside V.
- We want the evaluations of these descriptions to be reasonably absolute.
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- *Theories with constraints in Interpretation* (*TCIs*), and their models, are a formalisation of this idea.
- TCIs were used as a convenient means of defining state spaces of restricted abstract state machines.
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- TCIs provide bounds to the interpretation of a theory T over a signature σ.
- A TCI $\mathscr{T}$ expands on T by specifying additional requirements on a possible model $\mathfrak{M}$ of T.
- These requirements include
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Consistency of TCIs

Definition (L.)

A TCI is *consistent* iff it has a model in some outer model of V.

By examining a suitable forcing notion in V, we can decide if a TCI in V is consistent.

Proposition

Let $\mathscr{T}$ be a TCI. Then for all sufficiently large cardinals λ,

$$\Vdash_{Col(\omega,\lambda)} \exists \mathfrak{M} \text{ (“}\mathfrak{M} \textit{ is a model of } \mathscr{T}\text{”)} \iff \mathscr{T} \textit{ is consistent.}$$

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Local Method Definitions

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A *local method definition of* V is a non-empty class of TCIs definable in V.

- In the meta-theory, define a function Eval^V from the set of TCIs $\mathscr{T} \in V$ into the set of small extensions of V, such that

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If X and Y are local method definitions of V, $X \leq^M Y$ denotes the statement

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- Intuitively, $X \leq^M Y$ if V can see that Y provides non-trivial refinements to all consistent descriptions in X.
- $\leq^M$ partially orders local method definitions, so we can define the equivalence relation $\equiv^M$ the usual way.

Observation

Let X, Y be local method definitions. If $X \subset Y$, then $X \leq^M Y$.

Comparing Local Methods

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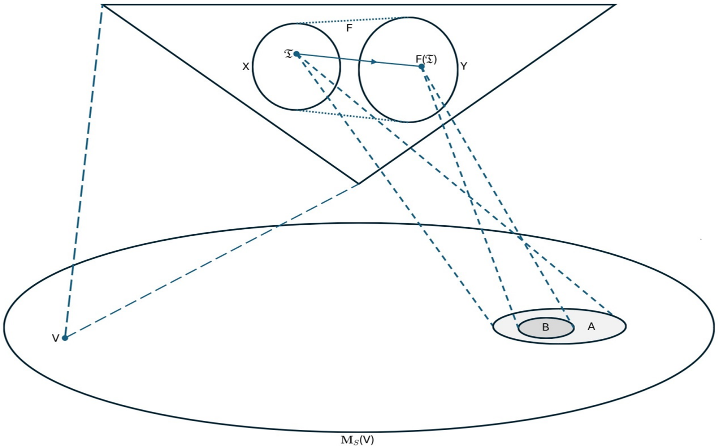

Figure: Visual representation of a function F witnessing $X \leq^M Y$, where X and Y are local method definitions.

The Local Method Hierarchy

- We can also define the complexity of a TCI by just looking at its first-order theory.
- For example, a Σ_n TCI has a first-order theory containing only Σ_n sentences.
- The classes of Σ_n and Π_n TCIs — denoted $\Sigma^{\mathsf{M}}_{\mathsf{n}}$ and $\Pi^{\mathsf{M}}_{\mathsf{n}}$ respectively — then form a hierarchy under the relation $\leq^M$, called the local method hierarchy.

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Theorem (L.)

Let $1 \leq n < \omega$. Then for every $\mathscr{T} \in \Pi^{\mathsf{M}}_{\mathsf{n+1}}$ there is $\mathscr{T}' \in \Sigma^{\mathsf{M}}_{\mathsf{n}}$ such that

$$\mathrm{Eval}^{V}(\mathscr{T}) = \mathrm{Eval}^{V}(\mathscr{T}').$$

Corollary

$\Pi^{\mathsf{M}}_{\mathsf{n+1}} \leq^{M} \Sigma^{\mathsf{M}}_{\mathsf{n}}$ *for all* $1 \leq n < \omega$.

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Set Forcing

- There is an obvious way of representing any forcing notion as a Π_2 TCI.
- Thus set forcing, as a technique of accessing small extensions of V, can be represented by a local method definition, denoted Fg.
- We want to see if Fg fits nicely in the local method hierarchy.
- From what we know so far, Fg $\leq^M \Sigma_1^{\mathrm{M}}$.

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Set Forcing is Σ_1 (is Π_2)

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$\mathrm{Fg} \equiv^M \Sigma_1^{\mathsf{M}}$.

Proof Idea.

By adapting our framework for constructing forcing notions to the context of TCIs, we associate with each Π_2 TCI $\mathscr{T}$ a forcing notion $\mathbb{P}(\mathscr{T})$, such that a unique model of $\mathscr{T}$ can be read off every $\mathbb{P}(\mathscr{T})$-generic filter over V.

This allows us to define a class function witnessing $\Pi_2^{\mathsf{M}} \leq^M \mathrm{Fg}$. $\square$

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Further Results

- We can refine the function witnessing $\Pi_2^{\mathrm{M}} \leq^M \mathrm{Fg}$ by iteratively pruning the $\mathbb{P}(\mathscr{T})$s of atoms.
- There are analogues of the previous theorem in V pertaining to certain countable Π_2 TCIs.
  - Essentially, we can define functions F of low complexity taking these TCIs to reals such that every $F(\mathscr{T})$-1-generic real codes a model of $\mathscr{T}$.

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Some Open Questions

(Q1) What structural properties of $(\mathbf{M}_F(L), \subset)$ also hold for $(\mathbf{M}_S(L), \subset)$?

(Q2) For example, is $(\mathbf{M}_S(L), \subset)$ downward directed?

(Q3) Are there $m, n < \omega$ for which $\Sigma^{\mathsf{M}}_{\mathsf{m}} \not\equiv^M \Sigma^{\mathsf{M}}_{\mathsf{n}}$?

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References I

[1] Gaisi Takeuti. "On the recursive functions of ordinal numbers". In: *Journal of the Mathematical Society of Japan* 12.2 (1960), pp. 119–128.

[2] Georg Kreisel and Gerald E Sacks. "Metarecursive sets". In: *The Journal of Symbolic Logic* 30.3 (1965), pp. 318–338.

[3] Dag Normann. "Set recursion". In: *Studies in Logic and the Foundations of Mathematics*. Vol. 94. Elsevier, 1978, pp. 303–320.

[4] Kurt Gödel. *Collected Works, Volume 2: Publications 1938-1974*. Oxford, England and New York, NY, USA, 1986.

References II

[5] Lenore Blum, Mike Shub, and Steve Smale. "On a theory of computation and complexity over the real numbers: NP-completeness, recursive functions and universal machines". In: *Bulletin of the American Mathematical Society* 21.1 (1989), pp. 1–46.

[6] Joel David Hamkins and Andy Lewis. "Infinite time Turing machines". In: *The Journal of Symbolic Logic* 65.2 (2000), pp. 567–604.

[7] Peter Koepke and Martin Koerwien. "Ordinal computations". In: *Mathematical Structures in Computer Science* 16.5 (2006), pp. 867–884.

[8] Wilfried Sieg. "Gödel on computability". In: *Philosophia Mathematica* 14.2 (2006), pp. 189–207.

References III

[9] Keng Meng Ng, Nazanin R Tavana, and Yue Yang. “A recursion theoretic foundation of computation over real numbers”. In: *Journal of Logic and Computation* 31.7 (July 2021), pp. 1660–1689. ISSN: 0955-792X. DOI: 10.1093/logcom/exab042. eprint: https://academic.oup.com/logcom/article-pdf/31/7/1660/40820416/exab042.pdf. URL: https://doi.org/10.1093/logcom/exab042.

Thank You!